

Thura Zaw

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WORK EXPERIENCE

General Magic (Thailand) Co., Ltd

Jun. 2025 – Mar. 2026

Robotic Engineer

Bangkok, TH

- General Magic Industries builds AI products, workflow automation, and blockchain-enabled intelligence for teams that want measurable leverage across operations, support, and product delivery.
- Designed and implemented the electrical system for a bipedal humanoid robot leg, enabling low-latency communication across 12 actuator joints using CAN Bus protocol and socketCAN on ROS2 control framework
- Developed custom PCBs and embedded firmware for actuator control utilizing tools such as KiCAD, LTSpice, STM32 Cube IDE, and STM32 Cube MX
- Collaborated within a team of 4 to achieve both sim-to-sim and sim-to-real prototyping in NVIDIA Isaac Sim, integrating commercial Cubemars actuators and 1 Mbps ISO CAN standard
- Coordinated efforts between university capstone requirements and company product goals, contributing technical expertise to a joint academic-industry robotics initiative

National Institute of Technology (Sendai College)

May 2025 – Jun. 2025

Robotics System Engineer Intern

Sendai, JP

- Engineering-focused technical college in Sendai, Japan, specializing in practical education, applied research, and interdisciplinary engineering training.
- Created an attention robot prototype using C++, Arduino, servo control, and OLED display, enabling real-time danger alerts for technicians through expressive actions and sound

EDUCATION

King Mongkut's Institute of Technology Lad Krabang (KMITL)

Apr. 2026

Bachelor of Robotics and Artificial Intelligence Engineering

Bangkok, TH

- **GPA: 3.7/4.0;** First Class Honor
- Mobile Robots, Introduction to Robotics & AI, Feedback Control, Kinematics & Dynamics, Computer Vision, Artificial Intelligence Technology, Microprocessor & Interface, Introduction to Algorithms, Differential Equations & Matrix Algebra, Probability & Statistics, Reinforcement Learning

PROJECTS

[Transformable Hybrid UAV-UGV System for Visual Inspection in Unstructured Environment](#)

- Led the design and implementation of a multi-joint robotic vehicle using ROS2 and SOTA vision system using Roboflow and custom trained AI Model for search and rescue mission after natural disasters.

[Leave Collector Robot with Adaptive Feedback Control Loop](#)

- Design a low-level control system for Ackermann Steering
- Develop a custom PCB for bridging high level ROS1 (Noetic) and low-level ATmega329P Microcontroller to ensure smooth actuator control and sensor response update.

VOLUNTEER EXPERIENCE

King Mongkut's Institute of Technology Ladkrabang (KMITL)

Jul. 2024 – Mar. 2025

Teaching Assistant (TA)

Bangkok, TH

- Assisted in designing and constructing laboratory experiments and demonstrations to enhance student understanding of complex concepts.
- Provided technical assistance and troubleshooting support for students during laboratory sessions, ensuring safe and effective use of electrical equipment.

CERTIFICATIONS, SKILLS, TECHNOLOGIES, LANGUAGES & INTERESTS

- **Certifications:** Altium Education PCB Design Course; Google Data Analytics (Coursera); Linux for Robotics (The Construct); C++ for Robotics (The Construct); Python for Robotics (The Construct)
- **Skills:** Sensor Fusion; Kalman Filtering; C Programming; ROS2; Robot Kinematics; SLAM (Simultaneous Localization and Mapping); ROS (Robot Operating System); Embedded Systems; Control Systems; Artificial Intelligence; C++; PyTorch; TensorFlow; Python Programming; Deep Learning; Computer Vision; Machine Learning
- **Technologies:** Altium Designer, SOLIDWORKS, Fusion360, KiCAD, LTSpice, Visual Studio, NVIDIA Isaac Sim, NVIDIA Isaac Lab
- **Languages:** English; Japanese
- **Interests:** Reading Novels, Play Guitar, Chess, Meditation, Gym Workout

ACHIEVEMENTS

K-Engineering World Tour 2024 Most Innovative Project Award

- Design a Cartesian Robot for drawing photorealistic image

YTU Rolympic Line Following Robot 2nd Prize

- Design a high speed line following robot using Arduino, IR sensors and PID control